DISCOVERIES & DIALOGUES

How do you reverse-engineer a star cluster?

Anna L. Rosen
San Diego State University



R136 · NASA, ESA, CSA, STScI

We measure starlight. We infer masses and ages.



Positions, colors, and brightnesses

Massive stars:
$M_\star \gtrsim 8\, M_\odot$

Rare, but powerful:
radiation, winds, and supernovae

Many faint stars remain unseen

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Predicting starlight. Inferring the physics.

Forward Model

What light would these stars produce?

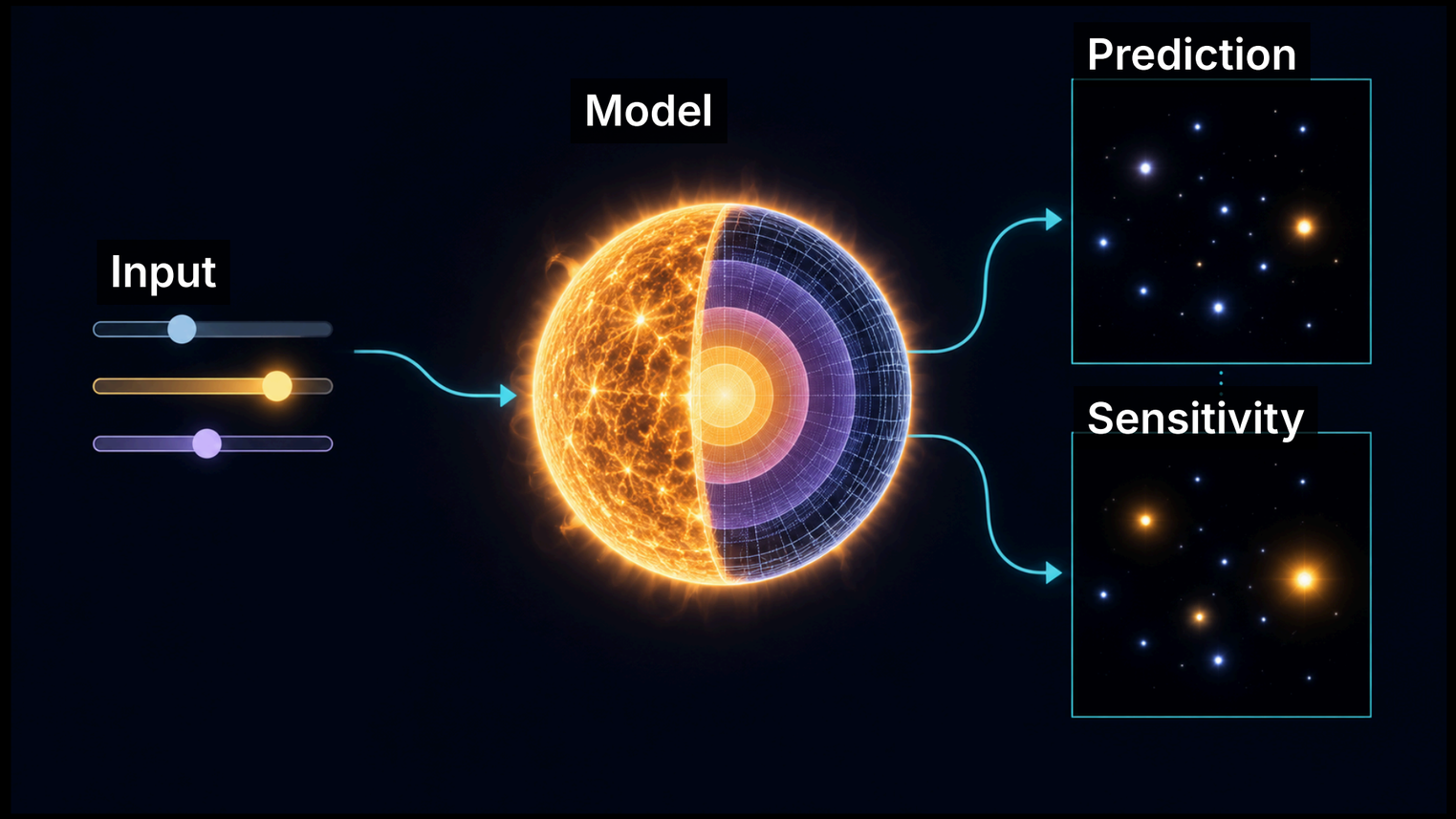
Inverse Problem

Which stars and physics could explain this light?

Different explanations can fit the same observations.

JAX computes predictions and their derivatives

A Python library for ML/AI and high-performance scientific computing.



JIT compilation
for CPUs, GPUs, and TPUs

Automatic differentiation
computes how predictions change with physical inputs.

Conceptual illustration · not simulation data

Jaxstro connects stellar physics to observations

Progenax
Birth populations

StarTrax
Rapid stellar evolution.

Stellax
1D stellar structure and evolution.

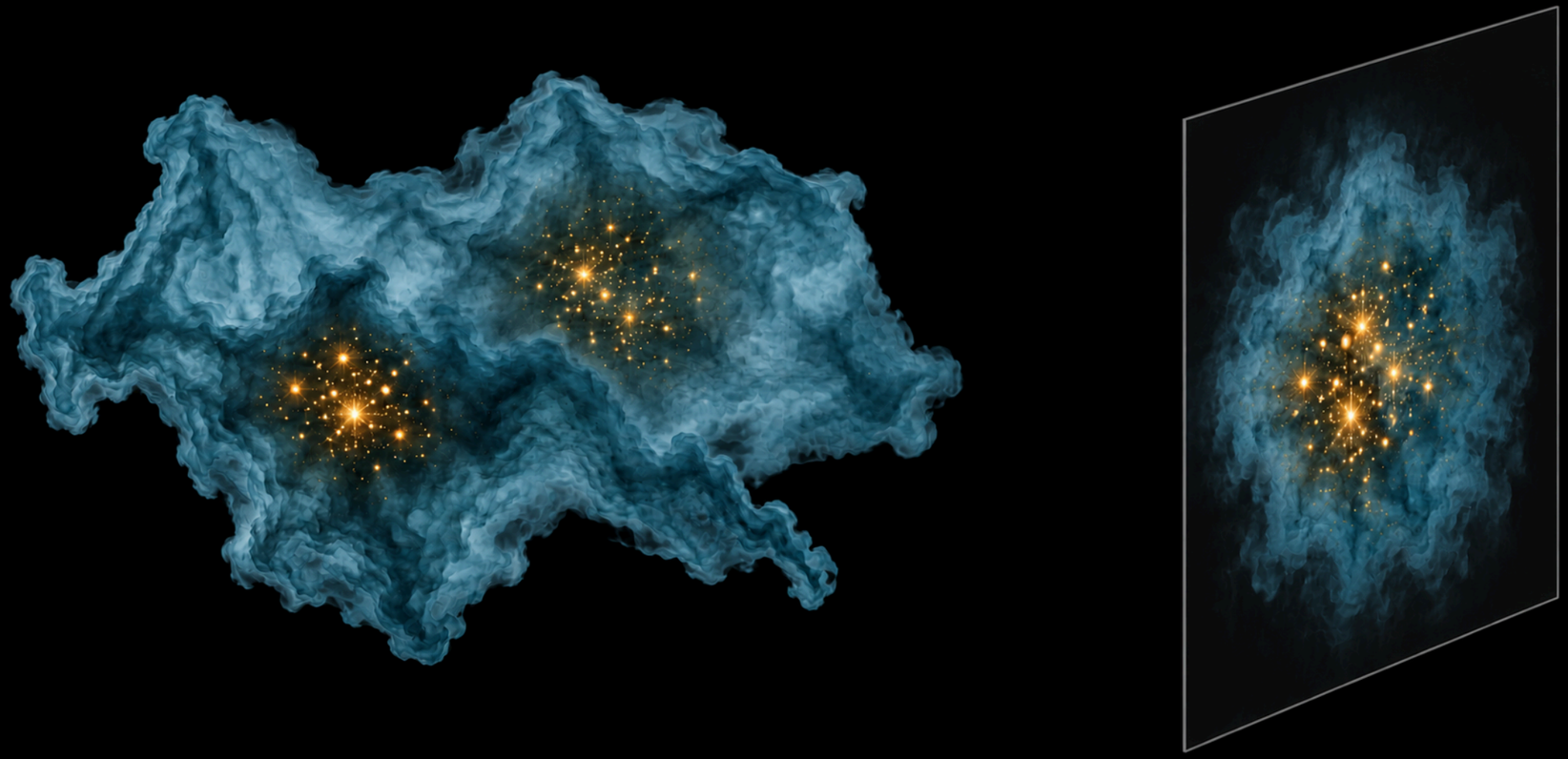
Gravax
Gravitational dynamics

Fluxax
Telescope observations

Informax
Inference

Jaxstro is a composable ecosystem designed for joint star-cluster modeling and inference.

Telescopes observe 2D projections of 3D clusters.

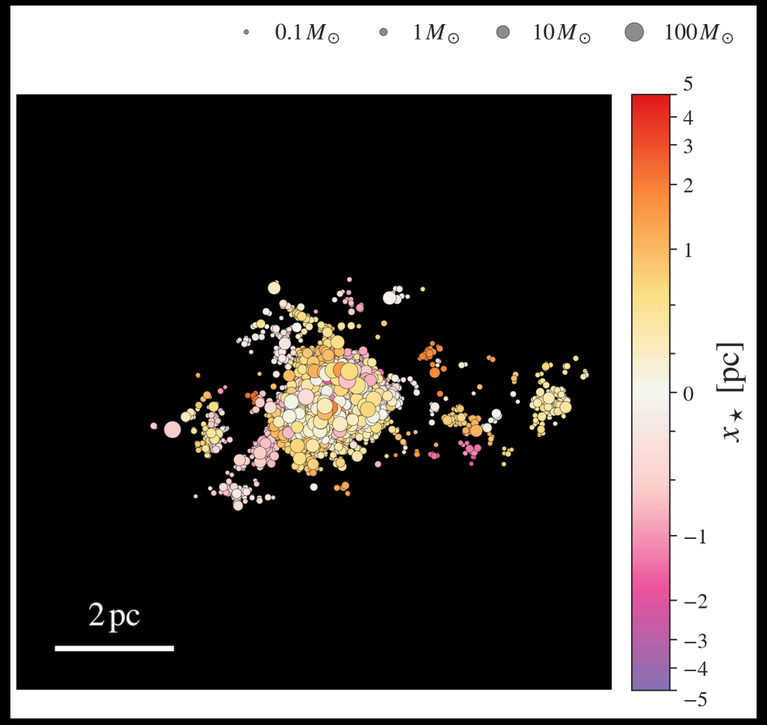


The same cluster looks different from different viewing angles.

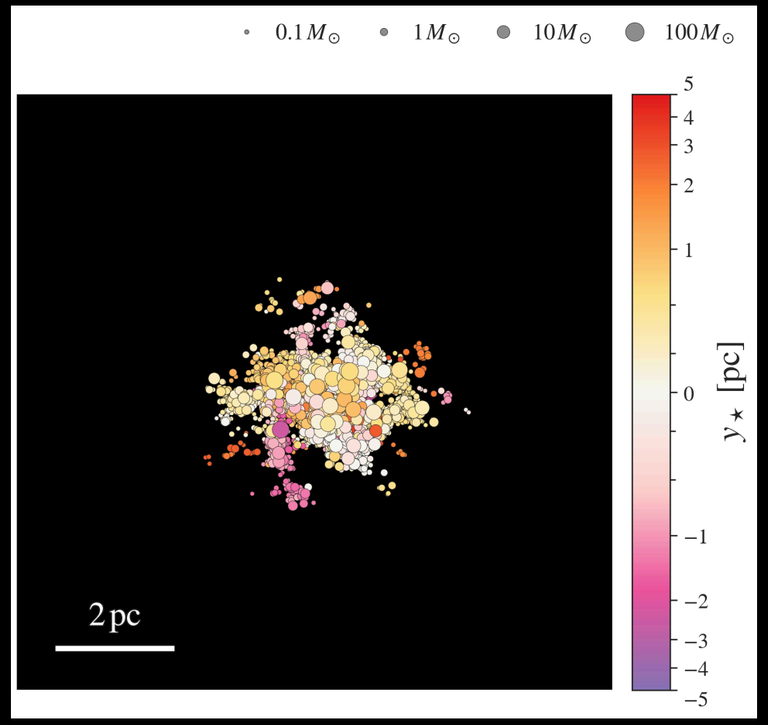
First view: OpenAI concept illustration. Second: saved Gravoturb / Progenax catalog · marker color encodes depth; size encodes mass

Telescopes observe 2D projections of 3D clusters.

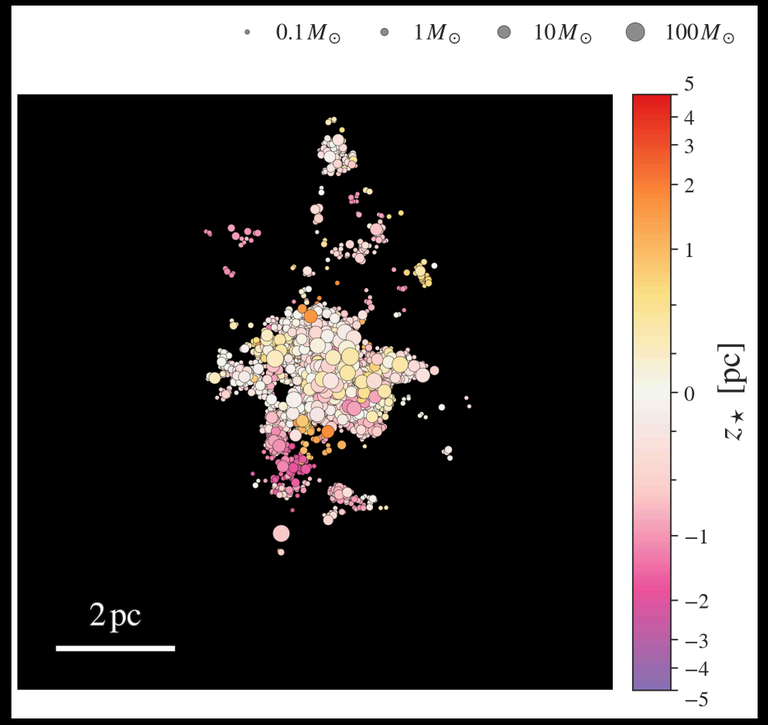
Same cluster. Three viewing directions.



Along x



Along y

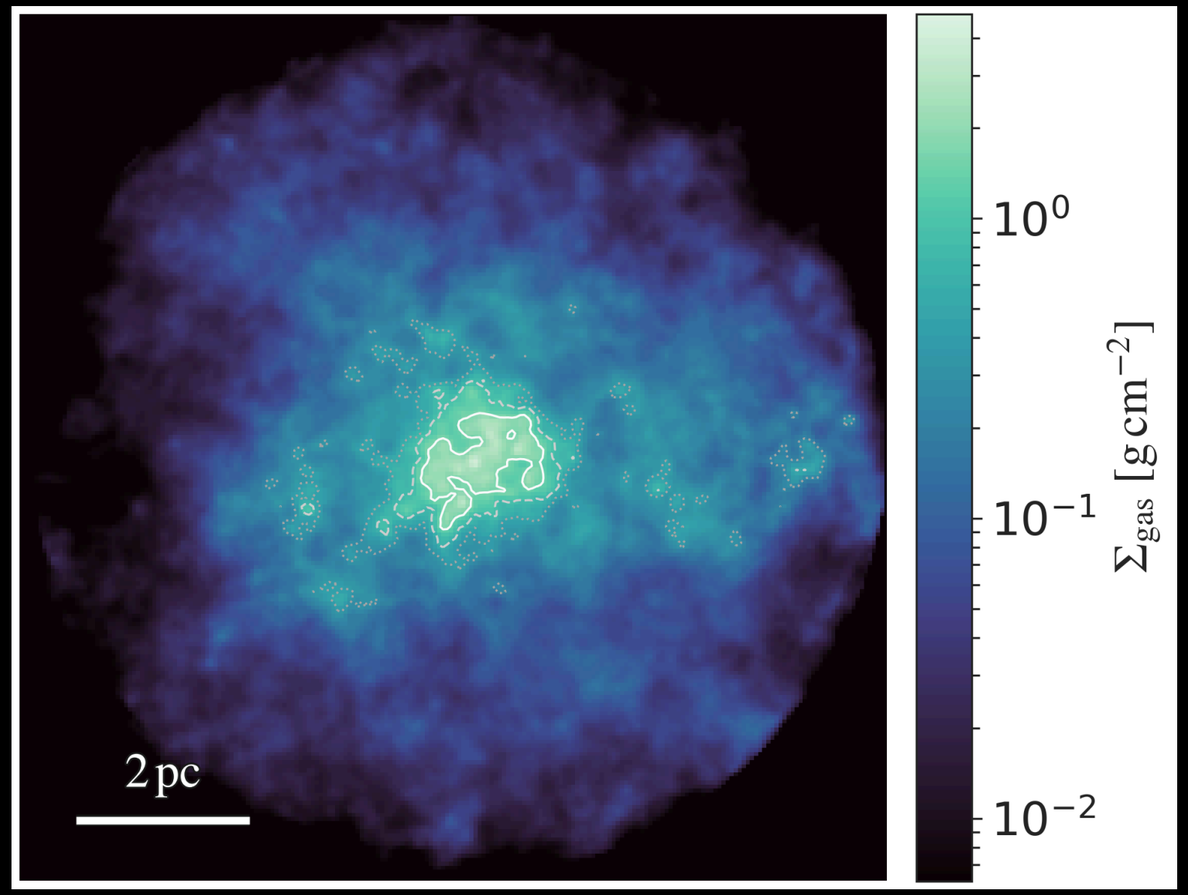


Along z

The same cluster looks different from different viewing angles.

First view: OpenAI concept illustration. Second: saved Gravoturb / Progenax catalog · marker color encodes depth; size encodes mass

Building a cluster from its birth cloud

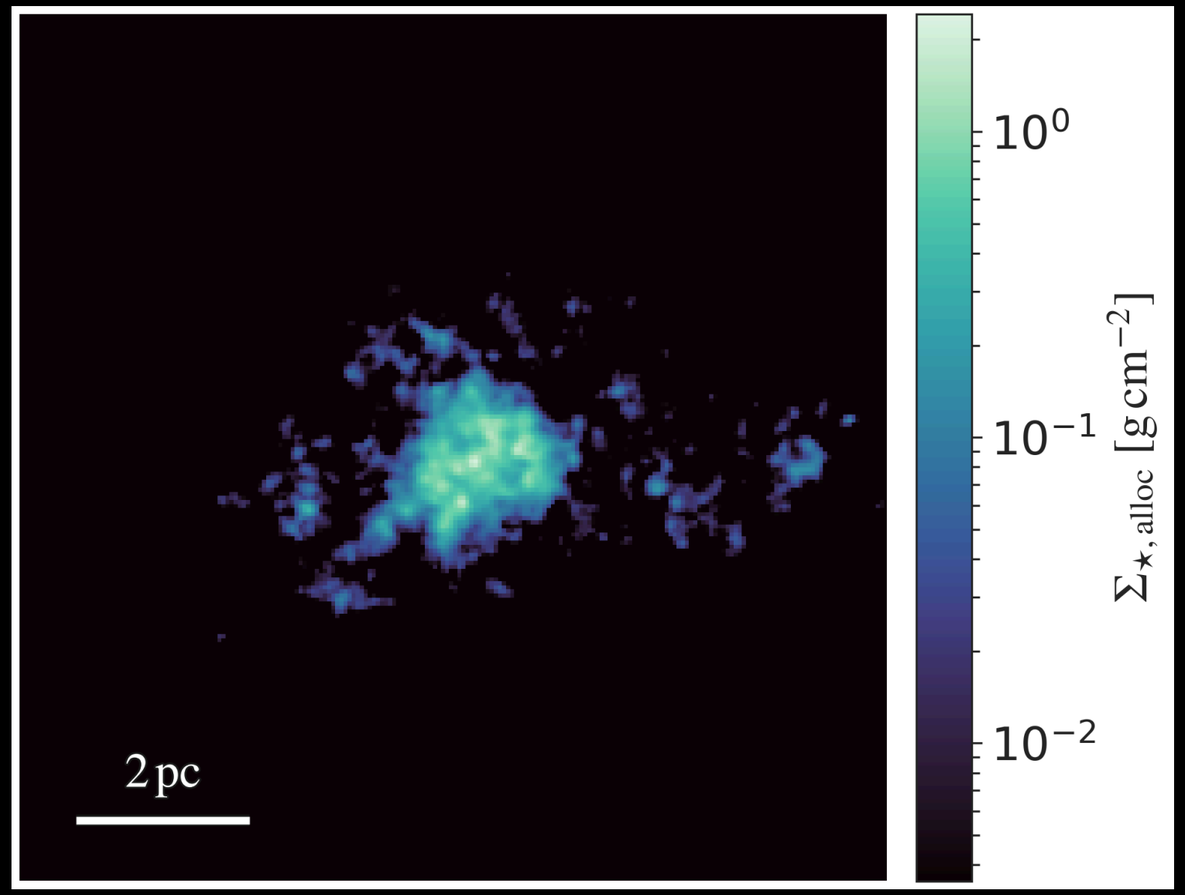


Denser gas collapses faster

Our model links gas density to where stars form

Gravoturb / Progenax · same example cluster · model construction

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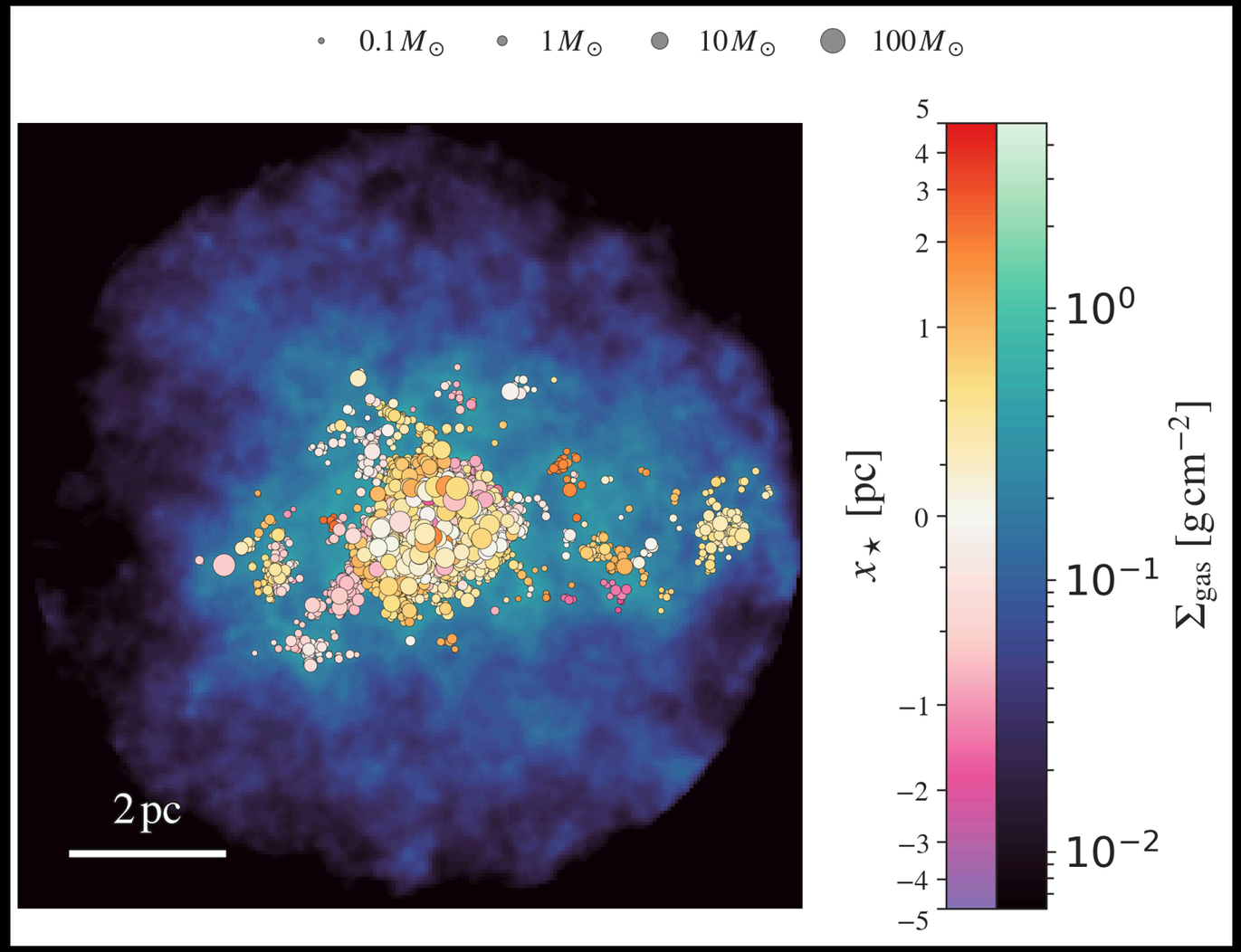


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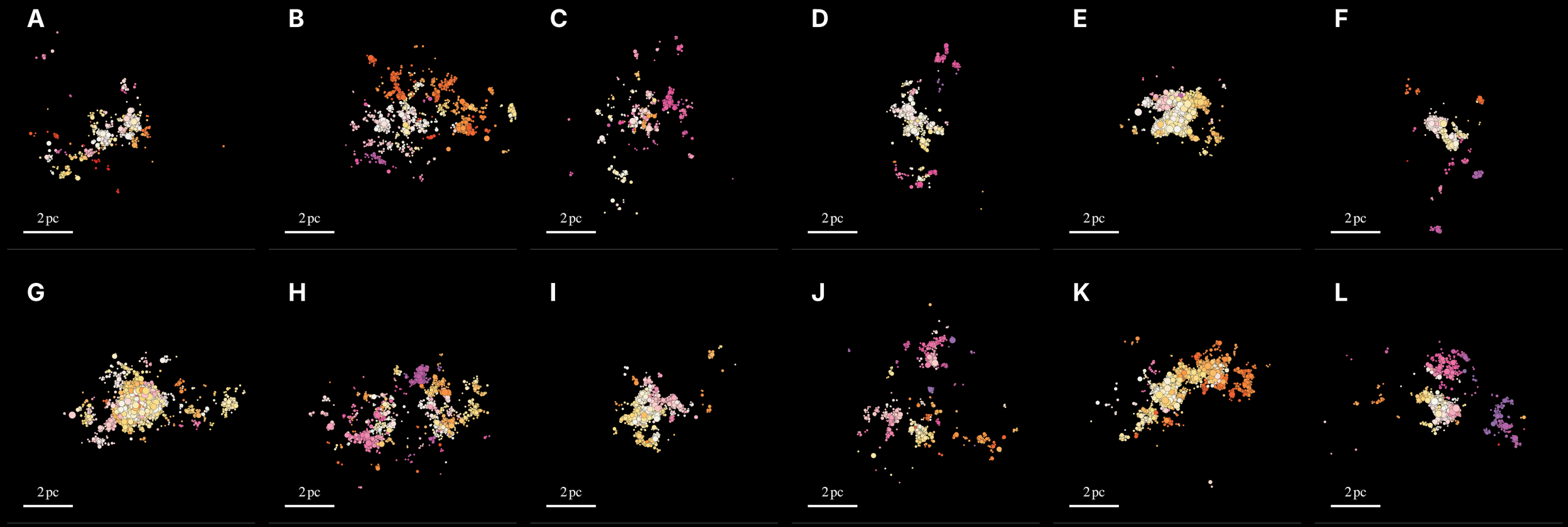


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Gravoturb / Progenax · same example cluster · model construction

One model produces many possible clusters

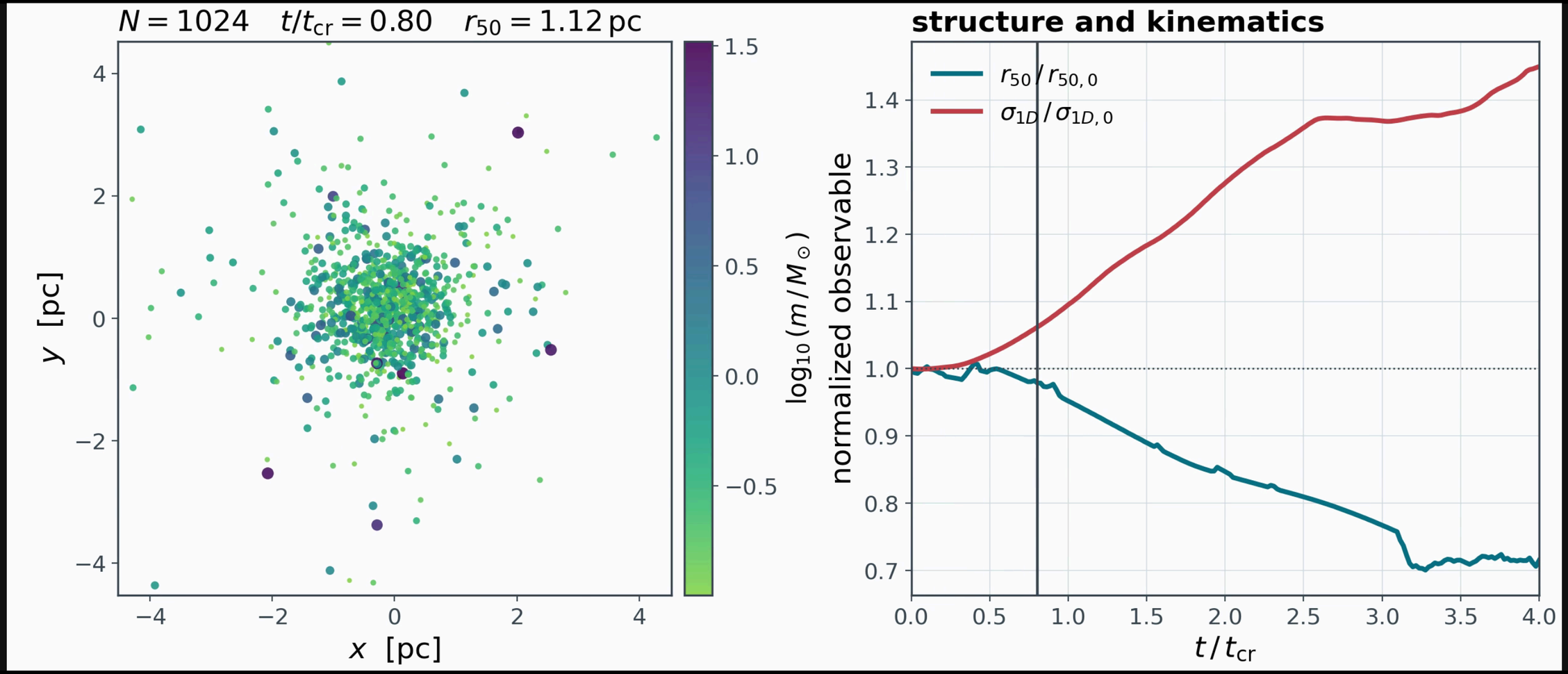


Same physical parameters. Independent random draws.

Marker size: stellar mass. Marker color: depth.

Twelve independent saved realizations · scale bars: $2\,\mathrm{pc}$

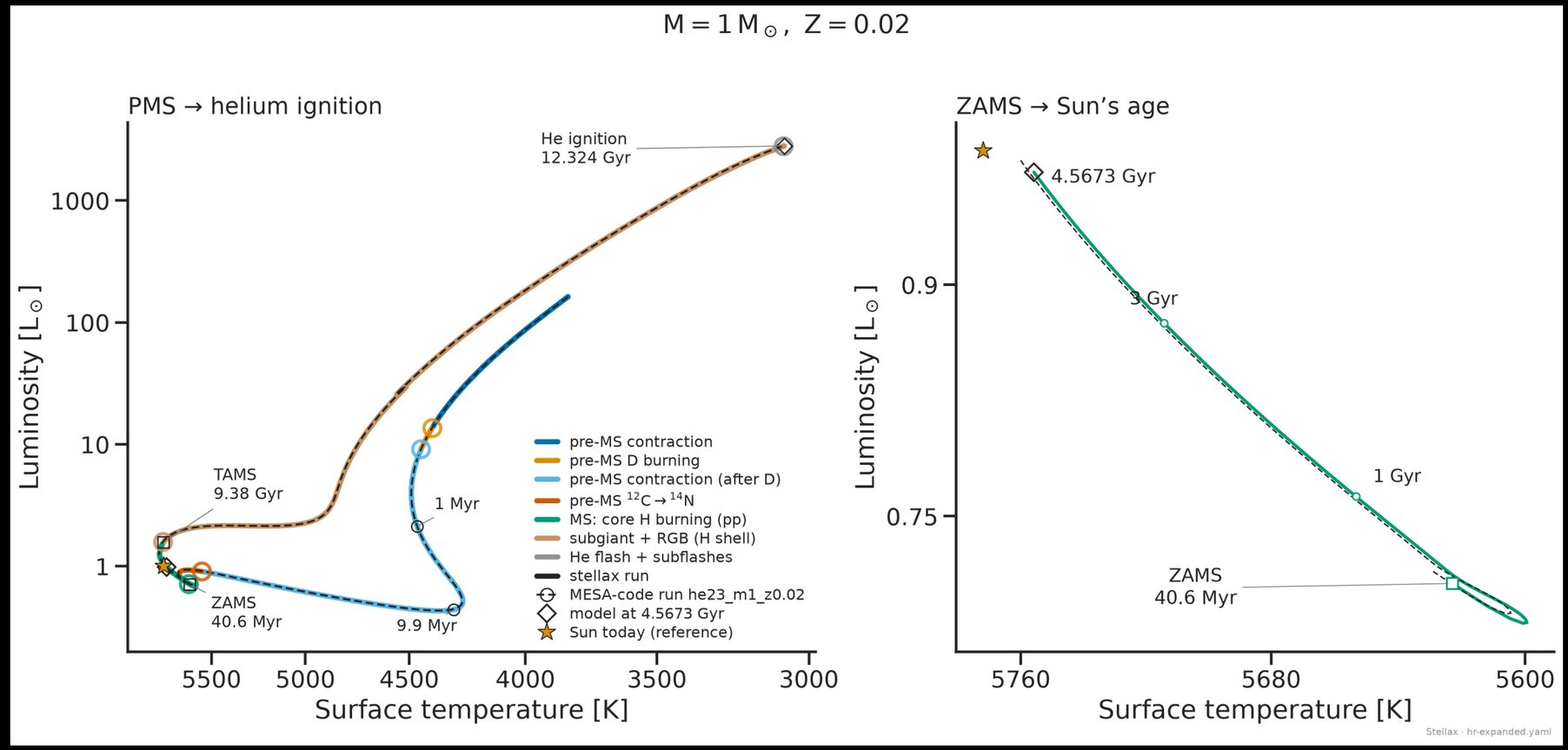
Gravity moves the stars after birth



A separate gravity-only simulation.

Stellar models connect physical inputs to observable light

Example: a Sun-like star · prediction



Predicting stellar evolution beyond the Sun's current age.

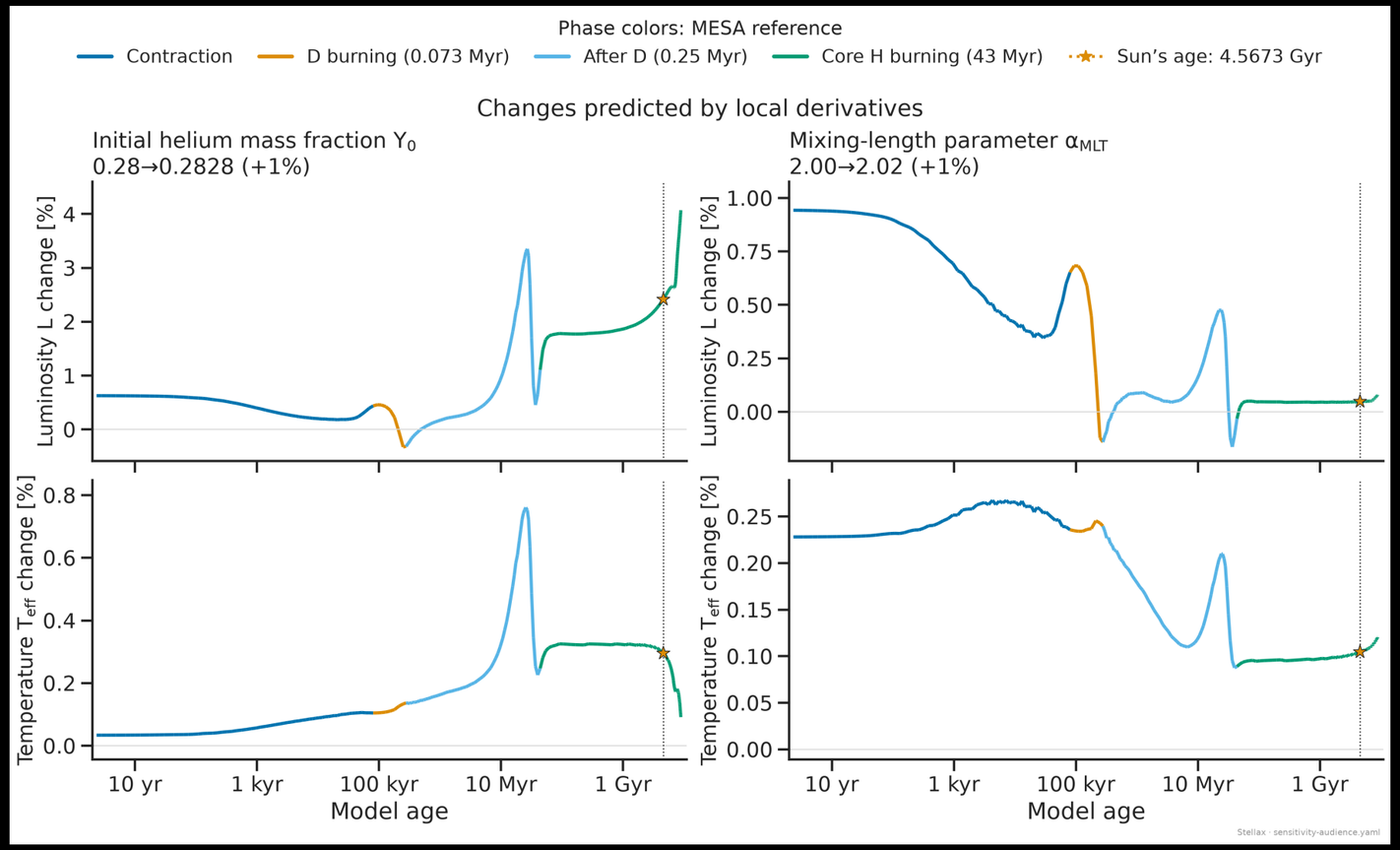
Broader calibration needs observations across mass, composition, and evolutionary stage.

Non-diffusive model · gold star: solar reference · idealized model clock from $\sim 2\,\mathrm{yr}$.

How do composition and energy transport affect stellar light?

Derivatives quantify how model inputs change the predictions.

How luminosity and temperature sensitivities change with age

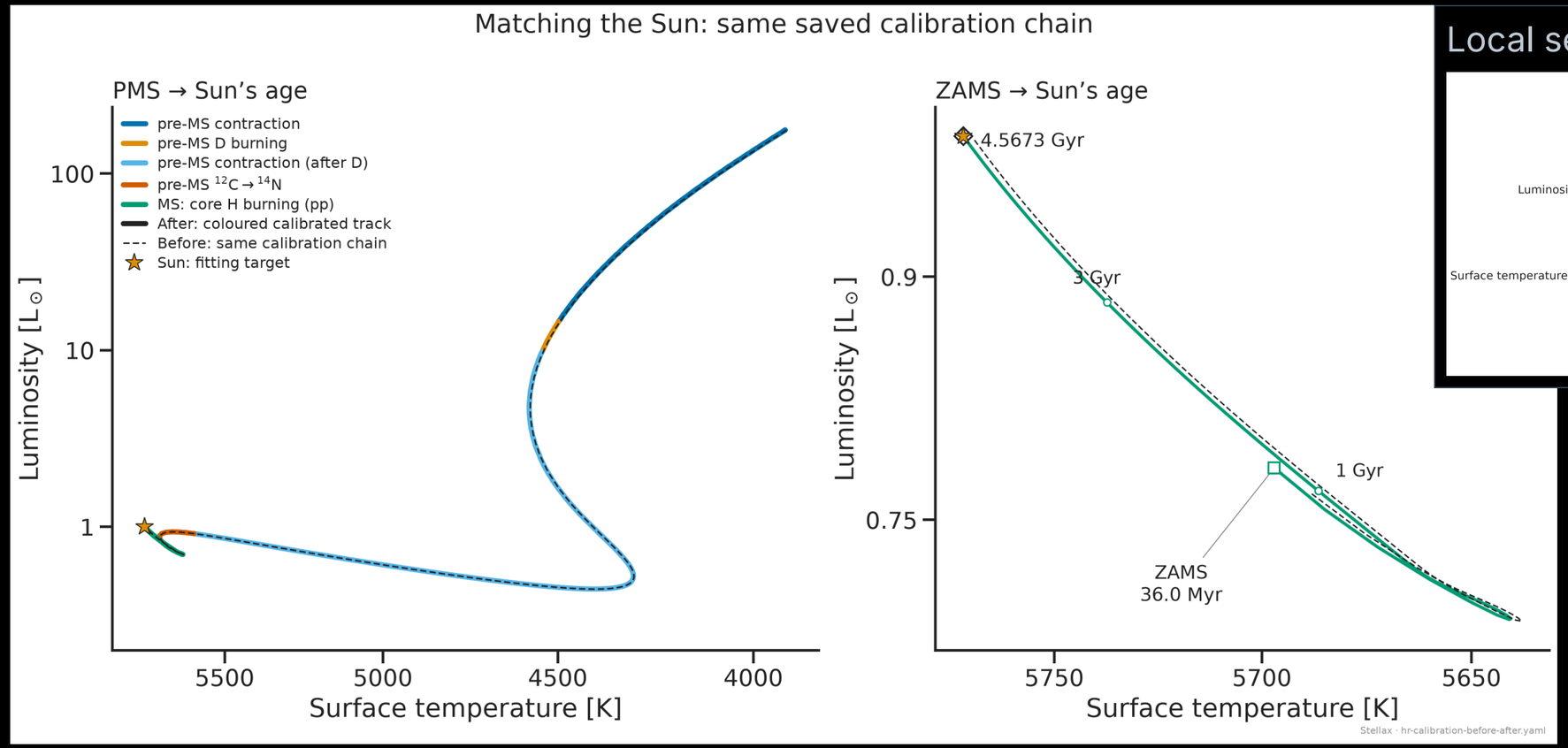


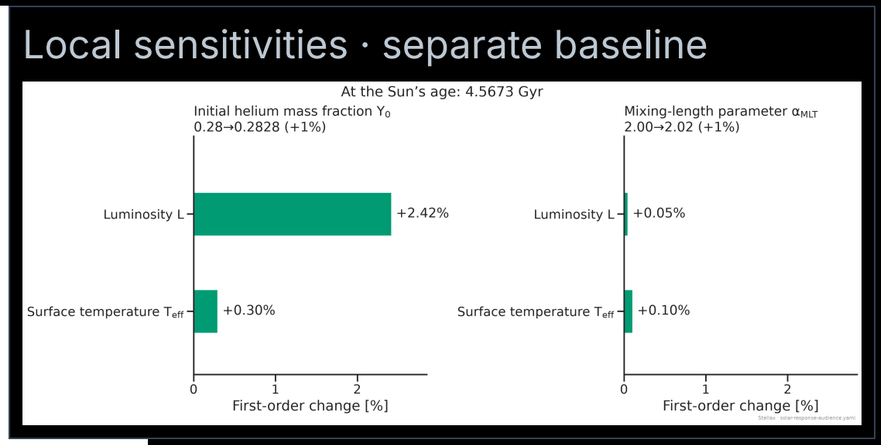
Local $+1\%$ relative input changes · fixed age/schedule · bars match the gold-star history values

Observations calibrate stellar models across masses and metallicities

Use measured stellar properties to constrain model inputs.

Example: calibration to solar observations.

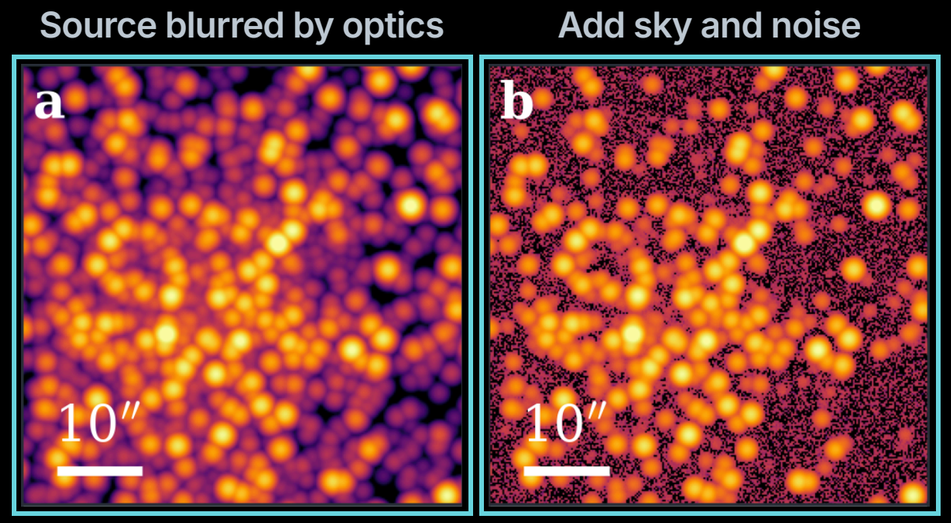




This fit tests one star. Other masses and metallicities require their own observational constraints.

Fixed $1\,M_\odot$, $4.5673\,\mathrm{Gyr}$ · fit L, R, and envelope Z/X with Y_0, Z_0, and α_{MLT} · own starting model and derivatives

Which model inputs change the predicted image?



ONE POPULATION · ONE OBSERVING MODEL

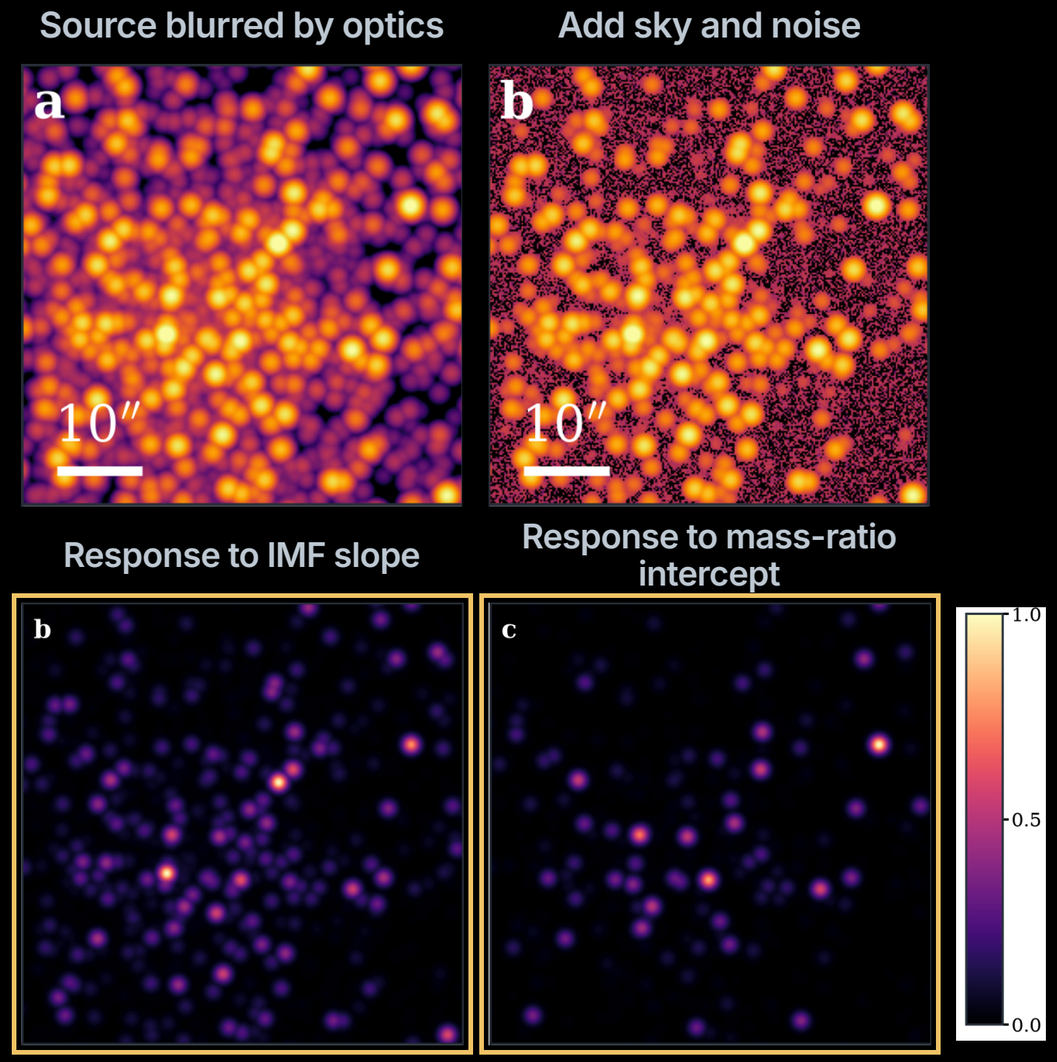
First, predict the telescope image.

IMF: the distribution of stellar birth masses

The top row keeps the stars fixed and adds sky and noise.

Fluxax · same saved population and field in both views · derivatives of the expected image

Which model inputs change the predicted image?



AUTOMATIC DIFFERENTIATION WITH JAX

Many inputs shape the light.

IMF slope

$$\left|\frac{\partial I_g}{\partial \alpha}\right|$$

Binary mass-ratio intercept

$$\left|\frac{\partial I_g}{\partial \gamma_0}\right|$$

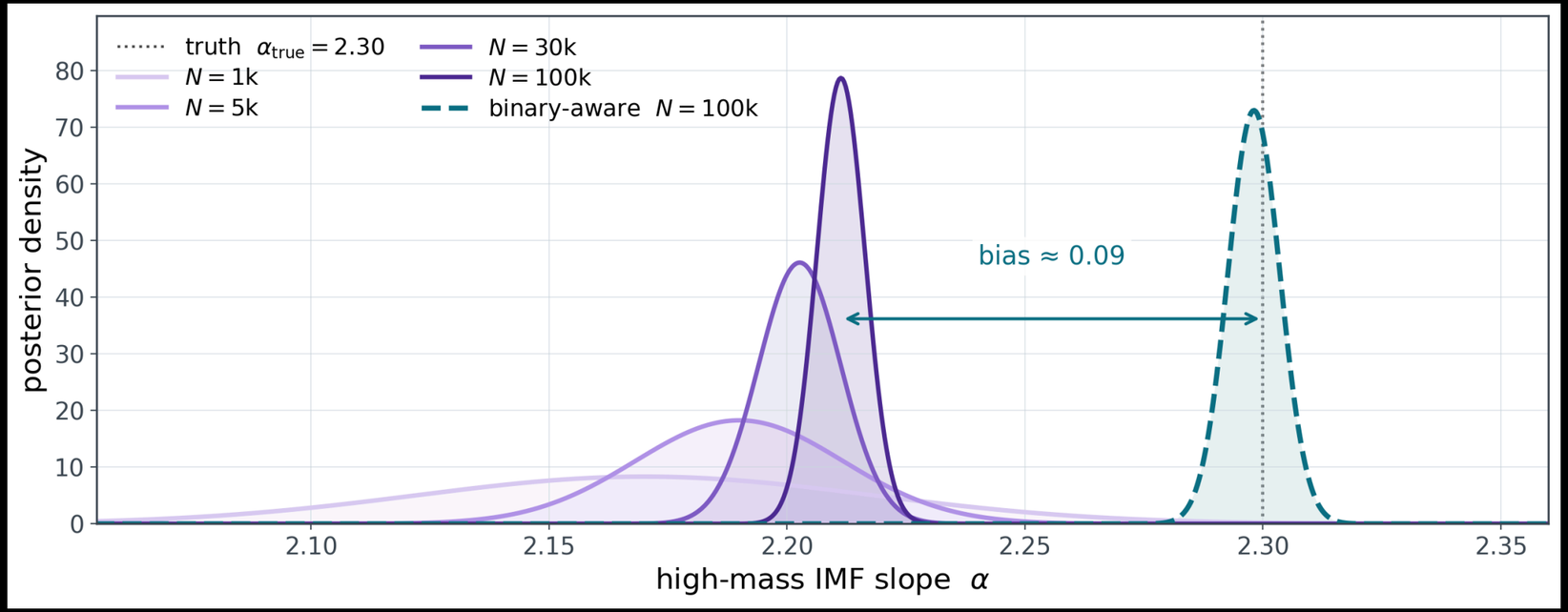
Bright pixels locate sensitivity to each input.

Each panel is normalized to its own peak.

Fluxax · same saved population and field in both views · derivatives of the expected image

More stars can make a wrong answer more precise

Idealized synthetic test



Ignore binaries:

uncertainty shrinks, bias remains

Include binaries:

recover the input slope in this controlled test

More data cannot fix a missing piece of the model.

Rosen (2026), *Confidently Wrong: Why Ignoring Binaries Biases IMF Inference at Large Sample Sizes*
The Astrophysical Journal, accepted · arXiv:2603.15779

Which observations can distinguish competing models?



Jaxstro connects stellar physics, cluster dynamics, and starlight to infer how star clusters form and evolve.

Motions
Constrain dynamics.

Spectra and companions
Constrain stellar properties and multiplicity.

R136 · NASA, ESA, CSA, STScI · proposed constraints, not a completed R136 inference

Modern astronomy is astrophysics

More than pretty space pictures.

Physics guides our models.
Observations test them.

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